# Chapter 6: Anatomy of Flowering Plants

The Tissue System

**BAIO LEARNING SYSTEM · Narendra Sir — Founder/CEO** Class Notes · Topic 1/8

**3 tissue systems: Epidermal, Ground, Vascular** **Stomatal apparatus = guard cells + subsidiary cells** **Trichomes (stem) vs Root hairs (root)** **Open (cambium) vs Closed vascular bundle** **Conjoint (stem/leaf) vs Radial (root)**

### *What is a Tissue System?*

A group of cells performing the same function is a **tissue**. A group of different tissues associated together to perform a function is a **tissue system**. Based on structure and location, plant tissue systems are of three types: **Epidermal**, **Ground (fundamental)** and **Vascular (conducting)**.

### *Epidermal Tissue System*

Forms the outermost covering of the plant body. Consists of **(a) epidermal cells** — elongated, compactly arranged parenchymatous cells with a large vacuole, forming a single-layered **epidermis** covered by a waxy **cuticle** (cuticle is absent in roots); **(b) stomata**; and **(c) epidermal appendages**.

### *Stomata & the Stomatal Apparatus*

Stomata are structures in the leaf epidermis that regulate **transpiration** and **gaseous exchange**. Each stoma is bounded by two bean-shaped **guard cells** (dumb-bell shaped in grasses) that control the opening/closing of the pore. Guard cells possess chloroplasts and are surrounded by specialized epidermal cells called **subsidiary cells**. The stomatal aperture, guard cells and subsidiary cells together form the **stomatal apparatus**.

### *Epidermal Appendages*

Epidermal cells bear hair-like structures. On the stem these are called **trichomes** — multicellular, branched or unbranched, and help reduce water loss during transpiration. On the root they are called **root hairs** — unicellular, and help absorb water and minerals. Some epidermal appendages may even be secretory.

### *Ground Tissue System*

All tissue except the epidermis and the vascular bundles belongs to the ground (fundamental) tissue system, made of simple permanent tissues — parenchyma, collenchyma or sclerenchyma. In primary roots and stems it is represented by the **cortex, pericycle, pith and medullary rays**; in leaves it is represented by the **mesophyll**.

### *Vascular Tissue System*

Consists of vascular bundles. **Xylem + Phloem together constitute a vascular bundle.** In dicotyledonous plants a couple of layers of meristematic cells lie between the xylem and phloem, called **cambium**, **vascular cambium** or **intrafascicular cambium**.

### *Open vs Closed Vascular Bundles*

A vascular bundle is **open** when cambium is present between the xylem and phloem; this cambium is capable of producing secondary growth, and open bundles are found in **dicot plants**. A vascular bundle is **closed** when cambium is absent between xylem and phloem, so no secondary growth is possible; closed bundles are found in **monocot plants**.

### *Conjoint vs Radial Arrangement*

When xylem and phloem lie on the **same radius** it is a **conjoint** arrangement, usually with phloem located outer to xylem; conjoint bundles are common in **stems and leaves**. When xylem and phloem lie on **alternating (different) radii** it is a **radial** arrangement, found in **roots**.

### Open vs Closed Vascular Bundle

| Feature | Open Vascular Bundle | Closed Vascular Bundle |
|---|---|---|
| Cambium | Present between xylem & phloem | Absent |
| Secondary growth | Possible | Not possible |
| Found in | Dicotyledonous plants | Monocotyledonous plants |

### The 3 Tissue Systems

- Epidermal tissue system — epidermal cells, stomata, appendages
- Ground (fundamental) tissue system — parenchyma/collenchyma/sclerenchyma
- Vascular (conducting) tissue system — xylem + phloem